

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of (B Sc) Biochemistry

Examination April-May 2019

Course code & Name: BC602 Plant Biochemistry

Date: 02/05/2019 Day: Thursday Time: 1:30 p.m To 3:30 p.m.

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

10

1. Precise role of reaction center in a photosystem is _____.
(i) capturing energy from the photons (ii) transferring an excited electron to the acceptor
(iii) that of a mobile electron carrier (iv) It serves as a storage for chlorophyll a
2. Which of the following processes occurs in stroma of the chloroplast?
(i) Calvin cycle (ii) Starch biosynthesis
(iii) Photolysis of water molecule (iv) Both (i) and (ii)
3. Putrescine is synthesized from _____.
(i) Lysine (ii) Arginine
(iii) Tryptophan (iv) Cadaverine
4. The ability of the callus to differentiate into a plant organ or a whole plant is regarded as _____.
(i) Dedifferentiation (ii) Plasticity
(iii) Recalcitrance (iv) Redifferentiation

5. Which of the following organelles are actively involved during nitrate assimilation in roots?
- (i) Vacuoles and Chloroplasts (ii) Vacuoles and Leuocoplasts
(iii) Vacuoles and Peroxisomes (iv) Vacuoles and Golgi apparatus
6. Special nitrogen-fixing organelles formed due to differentiation of *Rhizobium*, after infecting the legume host plant can be identified as _____.
- (i) nodule primordium (ii) infection thread
(iii) bacteroids (iv) None of the above
7. Which of the following enzymes cleaves from end of starch molecule to produce maltose?
- (i) Exoamylases (ii) Debranching enzyme
(iii) β -amylase (iv) α -amylase
8. Synthesis of which of the following carbohydrate molecules occurs on plasma membrane?
- (i) Cellulose (ii) Fructans
(iii) Sucrose (iv) Pectins
9. Flavonoids can be largely classified as _____.
- (i) alkaloids (ii) terpenes
(iii) cyanogenic glycosides (iv) phenolic compounds
10. During water translocation, negative water potential on the surface of the leaves is created due to _____.
- (i) poor water uptake (ii) closing of the stomata
(iii) photosynthesis (iv) transpiration

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Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q – II (A) Answer the following questions as directed (ANY TWO) (8)

- (i) Discuss the Z–diagram of photosynthesis and Calvin cycle.
- (ii) Discuss biosynthesis of starch and sucrose as main CO₂ assimilation products of plants
- (iii) Discuss compartmentalization of photorespiratory pathway.

(B) “Root cells contain more number of mitochondria as compared to leaf cells”. Justify the statement (2)

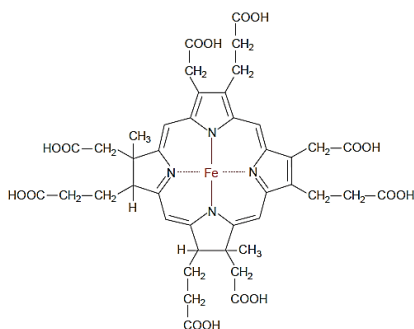
SECTION – II

Q-III (A) Answer the following questions as directed (Any FOUR) (12)

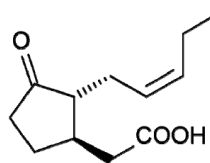
- (i) Explain nitrate assimilation in mesophyll and root cells.
- (ii) Describe nitrogenase complex and special structures for symbiotic nitrogen fixation.
- (iii) Give an account of any two types of plant secondary metabolites
- (iv) Enlist major applications of Plant Tissue Culture. Give an outline of Somatic embryogenesis
- (v) Give an account of macro and micronutrients important for plant growth

(B) Identify any three of the given structures and give their role. (3)

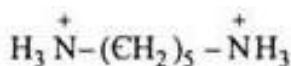
(i)



(ii)



(iii)



(iv)

